

GSOS URI Scheme

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1. Scope

The **gsos** URI scheme identifies a GSOS cell or a resource within it. It is intended for GSOS-capable clients and resolvers. A URI such as `gsos://K2/` identifies the root of standalone cell K2; it does not by itself guarantee network reachability, browser support, service availability, or authentication.

2. Syntax

The scheme follows the RFC 3986 generic URI syntax:

```
gsos-uri      = "gsos:" "/" namespace-id [ path-abempty ] [ "?" query ] [ "#"
fragment ]
namespace-id = 1*( ALPHA / DIGIT / "-" )
cell-segment = 1*( ALPHA / DIGIT / "-" )
```

ALPHA and DIGIT are the core rules defined by RFC 5234. `path-abempty`, `query`, and `fragment` are imported from RFC 3986.

The URI authority is a registered GSOS namespace-id with an explicit registry type: standalone cell or layer. For a standalone cell, the path identifies a resource inside it. For a layer, the first non-empty path segment is the cell-segment identifying a cell inside that layer; later segments identify resources within the cell. A client **MUST** determine the registered namespace type before interpreting the path. Registry records **MUST** prevent ambiguous reuse of an identifier as both types in the same resolution context.

Namespace and cell identifiers are case-insensitive for comparison; producers **SHOULD** emit canonical registered spelling. No userinfo or port is permitted. Resource paths are case-sensitive unless specified otherwise. An empty path and `/` both identify a standalone cell root; canonical output **SHOULD** use `/`. For a layer, the path `/` identifies its layer root, if supported, and not a cell. Percent-encoding follows RFC 3986; `query` and `fragment` do not select the namespace or cell.

Examples

```
gsos://K2/
gsos://K2/education
gsos://L2/D1/
gsos://L2/D1/education
```

3. Meaning and Resolution

A GSOS client parses the URI, validates the namespace and any layer cell segment, and asks a GSOS resolver for current service metadata for that cell. The resolver returns authenticated endpoint metadata and an expiry or version. The client selects an endpoint and then uses that endpoint's supported secure transport and application protocol. The URI is stable across endpoint

changes. A registry record identifies the controlling cell authority; an endpoint is not an identity.

GSOS cells including K2 and L2/D1 are implemented and operational through the current authenticated HTTPS-based GSOS routing environment. This deployed web access demonstrates the GSOS cell model, registry mapping, application routing, and live service operation. Native resolution of a gsos URI directly by an operating system or general-purpose browser is a separate interoperability layer. The lookup mechanism, trust anchors, signed metadata format, revocation, failover, and independent bootstrap mechanism for that native layer **MUST** be specified and verified before it is represented as a generally available independent GSOS resolver service.

4. Interoperability and Deployment

GSOS cells such as K2 and L2/D1 are currently accessible and operational through configured HTTPS web routes. An unconfigured browser or operating system may nevertheless not know how to open gsos URIs natively. Direct use of gsos URIs therefore requires an installed GSOS client, supported browser integration, a native protocol handler, or a configured web gateway.

IANA registration documents the scheme name but does not itself install a handler, configure DNS, or grant access to a service. A handler or gateway that maps a GSOS URI to an HTTPS endpoint **MUST** preserve the requested cell identity and validate the returned endpoint metadata.

5. Security and Privacy Considerations

Resolution data **MUST** be authenticated to prevent redirection to a malicious cell or endpoint. Clients **MUST** validate the cell authority and transport security and **MUST NOT** treat an arbitrary resolver response as authorization. Session credentials **MUST** be scoped to the cell and **MUST NOT** be copied into resolver queries or URI strings. Resolver requests may expose which cell a user visits; deployments **SHOULD** minimize logging and leakage. Clients require expiry, revocation, replay, and downgrade controls for cached metadata. A failed primary resolver **MUST NOT** silently fall back to an untrusted source.

GSOS URIs **MUST NOT** contain passwords, access tokens, private keys, session credentials, or personally identifiable information. Authenticated resolver metadata **MUST** bind the requested canonical GSOS identity to the returned endpoints, metadata version, expiry time, and issuing authority. Query and fragment components **MUST NOT** change namespace or cell identity and **MUST NOT** be treated as authorization credentials. Dot-segments and their percent-encoded equivalents **MUST** be normalized or rejected before resource authorization is evaluated.

6. Internationalization and Encoding

The scheme name, namespace-id, and cell-segment values are ASCII. Internationalized human-readable names **MAY** be provided as resolver metadata, but they **MUST NOT** be substituted for registered identifiers or used for identity, authorization, or security comparison. Other URI components use RFC 3986 percent-encoding rules.

7. Contact and Change Control

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